

Week 6 Activity Log

React Fundamentals & Final Project Briefing

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Internship Program: CoderAxo Web Development Internship

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Week Duration: July 20 – July 26, 2026

Week Overview

This week introduced React, and partway through, CoderAxo released the brief for the Final Project — the Project & Team Collaboration System, to be built on the mandatory MERN stack inside a strict 10-day window. The second half of the week split between finishing the React fundamentals and starting to plan that project seriously.

Objectives for the Week

- Understand React's component model well enough to build multi-component UIs
- Learn state, props, and side effects (useState/useEffect) properly, not just by copying examples
- Turn the Final Project brief into a realistic, scheduled build plan before writing any code

Daily Activity Breakdown

Mon, Jul 20

Session on React fundamentals — components, JSX syntax, and how the component tree maps to the rendered UI.

Tue, Jul 21

Covered props for passing data between components and the useState hook for local component state.

Wed, Jul 22

Learned the useEffect hook for side effects (data fetching, subscriptions) and the dependency array's role in when effects re-run.

Thu, Jul 23

Practiced building small component-based UIs, lifting state up between parent/child components, and basic controlled form inputs.

Fri, Jul 24

Introduced to React Router — routes, nested routes, and navigation without full page reloads.

Sat, Jul 25

Final Project guidelines received: Project & Team Collaboration System, mandatory MERN stack, 10-day build window, deadline August 5. Read through the full requirements document.

Sun, Jul 26

Planned the project — sketched the MongoDB schema (Users, Teams, Projects, Tasks), broke the brief into modules, and drafted a rough day-by-day build schedule before development starts.

Detailed Summary

React's mental model took a bit to adjust to after weeks of directly manipulating the DOM with vanilla JS — instead of finding an element and changing it, the idea is to describe what the UI should look like for a given state and let React handle the update. Once that clicked, particularly through the `useState/useEffect` exercises, building small stateful components felt much more natural. The bigger event of the week, though, was the Final Project brief landing on Saturday. Given the 10-day window and the number of required modules (Auth, Dashboard, Teams, Projects, Tasks, File Sharing, Notifications, plus security throughout), I spent Sunday almost entirely on planning — sketching the database schema and deciding to build it as a vertical slice, one complete module (backend + frontend together) at a time, rather than building every module's backend first and connecting the frontend later.

Tools & Technologies Used

- React 18
- Vite
- React Router
- MongoDB Compass (schema sketching)
- Draw.io (rough ER diagram planning)

Skills & Concepts Learned

- Building component-based UIs with props and state in React
- Managing side effects with `useEffect` and understanding dependency arrays
- Client-side routing with React Router
- Breaking a large, multi-module project brief into a realistic, time-boxed plan

Challenges Faced & How I Resolved Them

A recurring source of small bugs this week was forgetting dependencies in `useEffect`'s dependency array, which either caused an effect to run on every render or not to re-run when it should have. Reading through React's own guidance on the 'rules of hooks' cleared most of this up, but it's clearly a concept that needs deliberate attention rather than intuition.

Reflection

Getting the Final Project brief while still mid-way through learning React added some pressure, but it also gave the React sessions an immediate, concrete purpose — every hook and pattern learned this week was something I knew I'd need within days, which made the material stick better than usual.

Goals for Next Week

- Start Task 3 (a deployed React app) and begin active development on the Final Project's Authentication module
- Stick to the vertical-slice plan drafted this week rather than getting pulled into building every module halfway
- Keep the 10-day Final Project deadline (August 5) firmly in view